



Review Article

Air pollution and mortality among infant and children under five years: A systematic review and meta-analysis

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ABSTRACT

The main environmental cause of childhood mortality and morbidity around the world is related to air pollution. This meta-analysis aimed to examine the association between the mortality of infant and children under age of five and exposure to air pollution. PubMed/Medline, Web of Science, and EMBASE databases were searched to recognize studies published from 1992 to 2018. Twenty-seven studies met the inclusion criteria, of which eight studies were conducted in North America, eight in Europe, six in Asia, four in Latin-American and one in Africa. We used the OHAT method (Office of Health Assessment and Translation) for evaluation of the risk of bias. The pooled relative risks of the infant and child under-five mortality were estimated by Random and fixed-effect models. Increases in particulate matter (PM) concentration was related to infant and child under-five mortality ($PM_{2.5}$ 3.4%, per $10 \mu g/m^3$ (95% CI: 1.7%–5.4%); PM_{10} 2.5%, per $10 \mu g/m^3$ (95% CI: 1.6%–4.3%)). The percent change in the infant and child under-five mortality due to CO , NO_2 , SO_2 and O_3 exposure were 3.1% per 1 part per million (95% CI: 1.9–3.9%); 1.7% per 1 parts per billion (95% CI: 0.9–4.8%); 2.0% per 1 parts per billion (95% CI: 1.0–5.0%); and –1.1% per 1 parts per billion (95% CI: –2.6–0.3%), respectively. Conclusively, $PM_{2.5}$, PM_{10} , CO , NO_2 , and SO_2 were positively and significantly associated with both infant and child under-five mortality. We obtained clear evidence of an association between exposure to particulate and gaseous air pollutants with the infant and child mortality. Registration number in PROSPERO: CRD42019120816.

1. Introduction

Globally, ambient air pollution is a serious environmental and health problem, affecting both developed and developing countries especially in urban areas (Landrigan et al., 2018). Various developing countries derive energy from natural sources for commercial and industrial goals, but do not have sufficient technologies to diminish potential air pollution arising from these energy sources (Kurt et al., 2016). This polluted air breathed by 92% of the population across the worldwide (Adair-Rohani, 2018; Organization, 2016a). Moreover, 93% of all children live in environments with air pollution concentrations above the WHO guidelines (Adair-Rohani, 2018). Outdoor air pollution is a mixture of thousands of components. Among them, particulate matter includes $PM_{2.5}$ and PM_{10} (PM with aerodynamic diameter $\leq 2.5 \mu m$ and $10 \mu m$), and gaseous pollutants include carbon monoxide (CO), sulfur dioxide (SO_2), ground-level ozone (O_3) and nitrogen oxides, including nitrogen dioxide (NO_2) and nitrogen oxide (NO) are the most important air pollutants (Karimi and Samadi, 2018; Pope et al., 2018). These pollutants are primarily derived from direct

emissions from incomplete combustion processes such as diesel and gasoline fuel, industrial processes and coal burning for power generation (Schwarze et al., 2006).

Evidence obtained from experimental and epidemiologic studies support the association between air pollution and all-cause mortality, cardiopulmonary and respiratory mortality and hospital admission due to respiratory (e.g., asthma, chronic obstructive pulmonary disease, lung cancer) and cardiovascular diseases (e.g., myocardial infarction, heart failure, cerebrovascular accidents) (Cohen et al., 2017; Hoek et al., 2013; Karimi and Samadi, 2019; Mannucci et al., 2015). Furthermore, long-term exposure to air pollution could induce oxidative stress and inflammation, resulting in airway injury and dysfunction (Karimi and Samadi, 2019; Mannucci et al., 2015).

In susceptible population groups such as the elderly, pregnant women, people with cardiovascular and respiratory disorders, children, and infants are those most susceptible to the poor air quality (Mannucci et al., 2015). Due to the growth stage of infants (0–1 years of age) and children under five years, which makes them more vulnerable to air pollution (Adair-Rohani, 2018). This population is proportionally more

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exposed than adults to ambient air pollution due to higher ventilation rates, longer periods spent outdoors, and mouth breathing, that increase their intake of air pollutants (Gouveia et al., 2018). In 2016, ninety-three percent of the global population under 18 years of age were exposed to PM_{2.5} (Adair-Rohani, 2018). More than one in every four deaths of children under 5 years of age is associated with environmental risks (Organization, 2016b). Both ambient air pollution and household air pollution contribute to respiratory tract infections that resulted in 543 000 deaths in children under the age of 5 years in 2016 (Organization, 2016a).

Over the last three decades, robust relations between exposure to particulate and gaseous air pollutants with the infant and child under-five mortality are consistently reported in epidemiological studies (Bobak and Leon, 1992; Carbajal-Arroyo et al., 2011; Dales et al., 2004; Lelieveld et al., 2015; Litchfield et al., 2018; Loomis et al., 1999; Tanaka, 2015; Woodruff et al., 1997a; Yorifuji et al., 2016). Zeman. (1998) in a study report that the rate of deposition normalized to lung surface area, tended to be greater (35%) in the children versus the combined group of adults and adolescents for particles (Zeman, 1998). Salihi et al. (2011) report that the incidence of placental abruption can increase by exposure to PM (Salihi et al., 2011). In other studies, a significant relationship was observed between the infant and child under-five mortality with exposure to PM_{2.5}, NO₂, and O₃, but some results are not significant (Carbajal-Arroyo et al., 2011; Litchfield et al., 2018; Yorifuji et al., 2016). Additionally, fossil fuel combustion and traffic-related air pollution were associated with infant and child mortality (Lelieveld et al., 2015; Tanaka, 2015). However, a follow-up study conducted by Lipfert et al. (2000) indicates that infant mortality was not related to O₃ and CO exposure (Lipfert et al., 2000). Similarly finding in Canada show that child mortality was not associated with short-term exposure to SO₂ and NO₂ (Dales et al., 2004). Consequently, the present documents are inconsistent, and corroborating these findings by meta-analysis study is essential. This paper aimed to describe the association between infant (0–1 years of age) and child under-five years of age mortality with exposure to particulate and gaseous air pollutants by systematic review and meta-analysis.

2. Method

The protocol of this study has been previously registered in the International Prospective Register of Systematic Reviews (PROSPERO) and is available at <http://www.crd.york.ac.uk/PROSPERO/> under registration number CRD42019120816. A complete Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) checklist (Moher et al., 2015) can be found in the supplementary material (Tables S1).

2.1. Search methods

We searched PubMed/Medline, EMBASE, and Web of Science with no language and geographical restrictions for published paper from January 1, 1992 to December 30, 2018. The following search keywords were used: *air pollution, particles, particulate, PM_{2.5}, PM₁₀, carbon monoxide, CO, Sulfur dioxide, sulfur dioxide, SO₂, nitrogen dioxide, NO₂, nitric oxide, nitrogen oxide, nitrogen monoxide, NO, ozone, O₃* connected with *infant mortality, child mortality, fetal mortality, neonatal mortality*. Additionally, we manually searched the reference list of all the included articles as well as additional articles tool in PubMed to identify potentially relevant studies.

2.2. Inclusion and exclusion criteria

All original articles that investigated the exposure to ambient air pollutants, including PM_{2.5}, PM₁₀, CO, SO₂, O₃, and NO₂ with childhood mortality from birth to five years old were included in this pooled analysis. Since the effect size of infant and child mortality attributed to

air pollutants is negligible. Hence, we accepted all epidemiological and observational studies including case-control, cohort, cross-sectional, time-series, and ecological studies which provided a relative risk (RR), odds ratio (OR), regression coefficients and their 95% confidence interval (CI) or standard error (SE) or related data to compute this parameter. Furthermore, we excluded review articles, systematic review, summaries, toxicological and animal studies, editorial, case series, case reports, commentaries, and governmental reports. We also not accepted studies without clearly defining outcomes or no quantitative data.

2.3. Data extraction

Data extraction was manually undertaken by two investigators (B.K and B.S). Any disagreement in study selection and data extraction was resolved by further discussion. From the eligible studies, the following data were extracted: first author name, publication year, study year, sample size, study design, concentrations of air pollutant, province, country, confounder adjustments and main findings of the mortality. More details are given in supplementary Table S2, Appendix A.

2.4. Risk of bias assessment

The risk of bias for each included study was evaluated by the environmental risk of bias tool reported by the National Institutes of Environmental Health Sciences-National Toxicology program (NIEHS-NTP). Furthermore, eligibility for each study was determined by the Navigation Guide from the University of California, San Francisco (OHAT, 2015). The following criteria were used to evaluate the risks of bias for each selected study: selection bias, exposure assessment, outcome assessment, selective reporting, main confounders, incomplete outcome measurement, and conflict of interest (Table 2 and more systematically in Table S2, Appendix B and C). The risk of bias for the above criterion was separated to “low”, “probably low”, “probably high”, “high”, or “not applicable”.

2.5. Statistical methods

The pooled effect size was computed from relative risks (RR) and their 95% confidence intervals (CI) after standardized increment in pollutant concentrations as follow: 10 µg/m³ for PM₁₀ and PM_{2.5}, 1 part per million (ppm) for CO, 1 part per billion (ppb) for O₃, NO₂, and SO₂. Another reported unit was converted. In most of the studies, relative risks (RR) were reported. Another effect size was also converted to RR and then used for pooling the risk estimates. The following formula was applied for converting an odds ratio (OR) to a RR:

$$RR = OR / (1 - p_0 + (p_0 \times OR)) \quad (1)$$

Where p_0 is the risk of a positive outcome in the control or the unexposed group (Grant, 2014; Wang, 2013). Moreover, to convert odds into the baseline risk the following equation was used (Grant, 2014)

$$Risk = odds / (1 + odds) \quad (2)$$

We applied both random-effects and fixed-effects meta-analyses to synthesize and summarize the relative risk of infant and under-five mortality. We assessed heterogeneity of studies by P-value from the Chi-squared test and I² statistic. I² values of 75%, 50%, and 25% were considered as high, medium, and low heterogeneity. Random-effects model was employed, when P-value < 0.10 or I² > 50%. Conversely, when heterogeneity was absent, the fixed-effect model was used. Forest plots were used to present the distribution of obtained RR from each study (DerSimonian and Laird, 1986). The percent change (%) in the infant and child mortality per increment in pollutant concentrations are displayed as following equation (Nhung et al., 2017):

$$Percent\ change\% = (RR - 1) \times 100\% \quad (3)$$

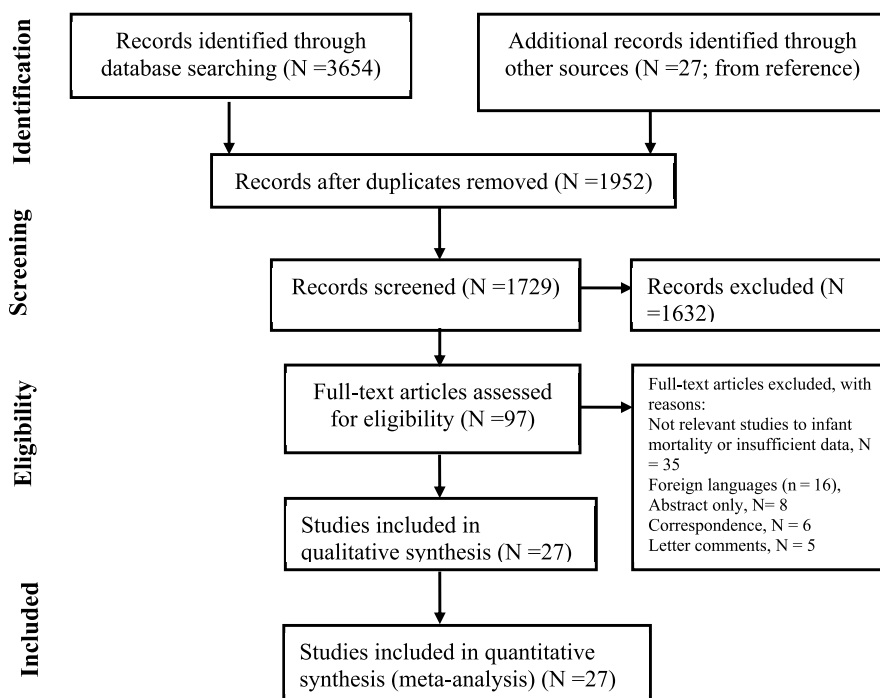


Fig. 1. PRISMA flowchart for study selection (Moher et al., 2015).

Egger's linear regression test was used to evaluate the publication bias. The statistical significance was obtained when P-value in Egger's test was less than 0.10. Publication bias was also examined by constructing Begg's funnel plots method and confirmed by Egger's method. Meta-regression and sensitivity analyses were used to recognize the reasons for the heterogeneity. The number of missing studies was computed by the trim-and-fill method. The statistical analyses were conducted by R Software (Version 3.2.2) and STATA Version 12.1 (STATA/SE, College Station, TX, USA). Significance for pooled estimates was defined as two-sided $P < 0.05$.

3. Results

3.1. Description of included studies

PRISMA Flowchart for study identification and literature selection (Moher et al., 2015) is given in Fig. 1. 3681 peer-reviewed articles were identified through literature search from the PubMed (2232 articles), ISI Web of Knowledge (854 articles), Embase (568 articles), and other sources (27 articles). Of these studies, 1952 articles were not relevant and excluded because of the duplication or overlap between databases. By considering the purpose of the study, 1632 papers were removed. The full texts of the 97 articles were completely read and after a qualitative assessment, 27 articles were finally accepted in our pooled analysis. Among them, the design of eight studies was case-crossover, five cohorts, four case-controls, four time-series, four cross-sections, and two ecological studies. Nineteen studies were published before 2010 (including 2010), while the others were published after 2010. Among the 27 selected studies, eight studies were conducted in North America, eight in Europe, six in Asia, four in Latin-American and one in Africa. The main characteristics of the included studies are summarized in Table 1 and more analytically in supplementary Table S2.

The risk of bias was very low in most of the selected studies (Table 2, Supplementary Table S2, Appendix B and C). Also, the risk of bias in some cases was 'probably low', and few studies had 'probably high' risk. None of our studies had a 'high' risk and therefore no studies were excluded from the analyses.

3.2. Concentrations of air pollutants

The pooled concentrations of pollutants were $46.5 \mu\text{g}/\text{m}^3$ for PM_{10} , $36.5 \mu\text{g}/\text{m}^3$ for $\text{PM}_{2.5}$, 70.0 ppb for NO_2 , 8.0 ppb for SO_2 , 29.0 ppb for O_3 , 3.5 ppm for CO , and 25.7 ppb for NO (Table 3, Supplementary Figs. S1–S8). The relative risks of mortality and related concentration of PM_{10} and $\text{PM}_{2.5}$ are shown in Fig. 2A and B.

3.3. Infant and child under-five mortality

The results obtained from the random-effect model show that infant and child mortality was positively associated with both $\text{PM}_{2.5}$ and PM_{10} . The pooled relative risk (RR) per each $10 \mu\text{g}/\text{m}^3$ increment in $\text{PM}_{2.5}$ and PM_{10} concentration were 1.034 (95% CI: 1.017 to 1.054) and 1.025 (95% CI: 1.016–1.43). These associations persisted, when RR was stratified by age (infant or child) or study design (Fig. 3A). Furthermore, all gaseous air pollutants, except ozone, were positively associated with infant and child under-five mortality. For example, carbon monoxide was positively associated with the infant and child mortality per 1 ppm increment (RR = 1.031 (95% CI: 1.019 to 1.039)) (Fig. 3B). Additionally, both sulfur dioxide (RR = 1.020 (1.00–1.05)) and nitrogen dioxide (RR = 1.017 (1.09–1.048)) were also positively associated with the infant and child mortality per 1 ppb increment. However, exposure to ozone per 1-ppb increment showed no association with the infant and child mortality (RR = 0.989, 95% CI: 0.974–1.003, $I^2 = 0.54\%$, $Q = 28.2$, $P = 0.08$). These associations persisted, when RR was stratified by age (infant or child) and study design (Fig. 3B, Supplementary Figs. S9–S10). The estimated RR of mortality was higher in children under-five years (1.021, 95% CI = 1.012–1.025, $I^2 = 72.8\%$, $p = 0.00$), compared to infants (1.015, 95% CI = 1.012–1.022, $I^2 = 55.0\%$, $p = 0.014$) (Supplementary Figs. S11). The percent change in the infant and child under-five mortality due to CO , NO_2 , SO_2 and O_3 exposure were 3.1% per 1 part per million (95% CI: 1.9–3.9%); 1.7% per 1 parts per billion (95% CI: 0.9–4.8%); 2.0% per 1 parts per billion (95% CI: 1.0–5.0%); and -1.1% per 1 parts per billion (95% CI: -2.6 – 0.3%), respectively. The highest percent change for child under-five mortality was related to CO and then SO_2 . Fig. 4A

Table 1
Included studies in the meta-analysis.

#	Author and year	Location	Type of study	Number of cases/ controls	Pollutants	Follow up period	Study population	References
1	Lipfert et al. (2000)	USA	Cohort	38 013/–	PM ₁₀ , CO, SO ₂ , NSPM ₁₀	1990	Infant	Lipfert et al. (2000)
2	Jakubiak-Lasocka et al. (2014)	Warsaw, Poland	Cross-Sectional	6375/–	PM ₁₀ , PM _{2.5}	2011–2013	Children and Infants	Jakubiak-Lasocka et al. (2014)
3	Litchfield et al. (2018)	UK	Case-Crossover	211/–	PM ₁₀ , SO ₂ , CO, O ₃ , NO ₂ , NO, NOx	1996–2006	Infant	Litchfield et al. (2018)
4	Gouveia et al. (2018)	Mexico City, Santiago, Chile, Sao Paulo, Rio de Janeiro	Time-Series	8762/–	PM ₁₀ , O ₃	1997–2005	Children and Infant	Gouveia et al. (2018)
5	Ha et al. (2003)	Seoul, South Korea	Cross-Sectional	1045/–	PM ₁₀ , CO, NO ₂ , SO ₂ , O ₃	1995–1999	Infant	Ha et al. (2003)
6	Yang et al. (2006)	Taipei, Taiwan	Case-Crossover	471/–	PM ₁₀ , SO ₂ , NO ₂ , CO, O ₃	1994–2000	Infant	Yang et al. (2006)
7	Ritz et al. (2006)	Southern California, USA	Case-Control	13 146/151 015	CO, NO ₂	1989–2000	Infant	Ritz et al. (2006)
8	Son et al. (2008)	Seoul, Korea	Case-Crossover	9137/–	O ₃ , PM ₁₀	1999–2003	Neonatal and Post-neonatal	Son et al. (2008)
9	Carbajal-Arroyo et al. (2011)	Mexico, Latin-American	Case-Crossover	12 079/–	PM ₁₀ , O ₃	1997–2005	Infant	Carbajal-Arroyo et al. (2011)
10	Woodruff et al. (2007)	USA	Cohort	6639/–	PM ₁₀ , PM _{2.5} , CO, SO ₂ , O ₃	1999–2002	Infant	Woodruff et al. (2007)
11	Egondi et al. (2018)	Korogochi and Viwandani in Nairobi, Kenya	Cohort	4529/–	PM _{2.5}	2003–2013	Children aged below 5 years	Egondi et al. (2018)
12	Kaiser et al. (2004)	25 counties, USA	Cohort	700 000/–	PM ₁₀	1989–1991	Infants	Kaiser et al. (2004)
13	Scheers et al. (2011)	Brussels, Belgium	Case-Crossover	2382/–	PM ₁₀	1998–2006	Infant mortality	Scheers et al. (2011)
14	Diaz et al. (2004)	Madrid, Spain	Cross-Sectional	–/–	TSP, SO ₂ , O ₃ , NOx	1986–1997	Children younger than 5 years	Diaz et al. (2004)
15	Romieu et al. (2004)	Ciudad Juarez, Mexico, Latin-American	Case-Crossover	628/–	PM ₁₀ , O ₃	1997–2001	Infant	Romieu et al. (2004)
16	Al-Hemoud et al. (2018)	urban districts in Kuwait	Cross-Sectional	235/–	PM ₁₀ , PM _{2.5}	2014–2016	Postneonatal Infant	Al-Hemoud et al. (2018)
17	Conceicao et al. (2001)	Sao Paulo, Brazil	Time-Series	13.1/–	PM ₁₀ , SO ₂ , CO, O ₃	1994–1997	Children under 5 years of age	Conceicao et al. (2001)
18	Woodruff et al. (2006)	California, USA	Case-Control	788/3089	PM _{2.5}	1999–2000	Infants	Woodruff et al. (2006)
19	Woodruff et al. (1997a, 1997b)	86 metropolitan in USA	Cohort	3 788 079/–	PM ₁₀	1989–1991	Infants	Woodruff et al. (1997a, 1997b)
20	Bobak and Leon (1999b)	Czechoslovakia, Czech Republic	Ecological study	971/–	TSP, SO ₂ , NO ₂	1986–1988	Infant	Bobak and Leon (1999b)
21	Tsai et al. (2006)	Kaohsiung, Taiwan	Case-Crossover	207/–	PM ₁₀ , SO ₂ , NO ₂ , CO, O ₃	1994–2000	Infant	Tsai et al. (2006)
22	Gouveia and Fletcher (2000)	Sao Paulo, Brazil	Time-Series	151 756/–	PM ₁₀ , SO ₂ , NO ₂ , O ₃ , CO	1991–1993	Infant	Gouveia and Fletcher (2000)
23	Hajat et al. (2007)	10 English cities, UK	Time-Series	22 288/–	CO, NO, NO ₂ , O ₃ , PM ₁₀ , SO ₂	1990–2000	Infant	Hajat et al. (2007)
24	Yorifuji et al. (2016)	Japan	Case-Crossover	2086/–	PM _{2.5} , PM ₇ , PM _{2.5-7}	2002–2013	Infants	Yorifuji et al. (2016)
25	Bobak and Leon (1999a)	Czechoslovakia, Czech Republic	Case-Control	2494/28 639	TSP ₁₀ , SO ₂ , NOx	1989–1991	Infant	Bobak and Leon (1999a)
26	Bobak and Leon (1992)	Czechoslovakia, Czech Republic	Ecological study	2699/–	TSP ₁₀ , SO ₂ , NOx	1986–1988	Infant	Bobak and Leon (1992)
27	Klonoff-Cohen et al. (2005)	California, USA	Case-Control	169/169	CO, NO ₂	1980–1999	Infant	Klonoff-Cohen et al. (2005)

Table 2
Risk of bias for each included study.

#	Study	Exposure assessment	Outcome assessment	Confounding	Selection bias	Incomplete outcome data	Selective outcome reporting	Conflict of interest	Other
1	Lipfert et al. 2000								
2	Lasocka et al. 2014								
3	Litchfield et al. 2018								
4	Gouveia et al. 2018								
5	Ha et al. 2003								
6	Yang et al. 2006								
7	Ritz et al. 2006								
8	Son et al. 2008								
9	Arroyo et al. 2011								
10	Woodruff et al. 2007								
11	Egondi et al. 2018								
12	Kaiser et al. 2004								
13	Scheers et al. 2011								
14	Diaz et al. 2004								
15	Romieu et al. 2004								
16	Al-Hemoud et al. 2018								
17	Conceicao et al. 2001								
18	Woodruff et al. 2006								
19	Woodruff et al. 1997								
20	Bobak and Leon 1999								
21	Tsai et al. 2006								
22	Gouveia et al. 2000								
23	Hajat et al. 2007								
24	Yorifuji et al. 2016								
25	Bobak and Leon 1999								
26	Bobak and Leon 1992								
27	Cohen et al. 2005								
28	Risk of bias rating	Low		Probably low		Probably high		High	

Table 3
The pooled concentration and descriptive statistics of air pollutant.

Pollutants	Mean	S.D	Median	S.E	Min	Max	I-squared	P-value
PM ₁₀ (µg/m ³)	46.4	18.8	4.2	37.8	4.0	1016.4	48.10%	0.00
PM _{2.5} (µg/m ³)	36.6	29.0	23.1	11.8	8.4	483.6	97.70%	0.00
TSP (µg/m ³)	56.1	19.5	12.5	56.1	33.5	115.5	61.10%	0.00
NO ₂ (µg/m ³)	143.5	13.8	13.9	23.9	49.0	1742.5	95.30%	0.00
SO ₂ (µg/m ³)	22.6	4.8	12.0	6.9	19.7	1557.6	62.30%	0.00
CO (µg/m ³)	4000.0	6.9	500.0	6.1	1375.0	86375.0	83.00%	0.00
O ₃ (µg/m ³)	62.3	16.9	3.9	22.9	49.0	470.2	71.10%	0.00
NO _x (µg/m ³)	131.3	73.4	32.3	64.5	86.4	762.7	0.00%	0.802
NO (µg/m ³)	34.4	19.0	5.2	28.0	37.5	337.5	93.80%	0.00

and B shows the pooled percent changes (%) of infant and child under 5-years mortality with exposure to the air pollutants and design of the study.

3.4. Additional analyses

Among all air pollutants, heterogeneity was observed. The lowest

heterogeneity was related to SO₂ ($I^2 = 46.2\%$). But, PM₁₀ was associated with the highest heterogeneity ($I^2 = 81\%$). A funnel plot was slightly asymmetric for PM₁₀ and SO₂ (Supplementary Figs. S13–S16). For PM₁₀, the shape of the funnel plot was slightly asymmetrical, and the Egger's regression intercept was $t = 3.53$ and $p = 0.04$, suggesting that there was possible publication bias among the 27 included studies. The result of the Trim and fill method show that approximately forty

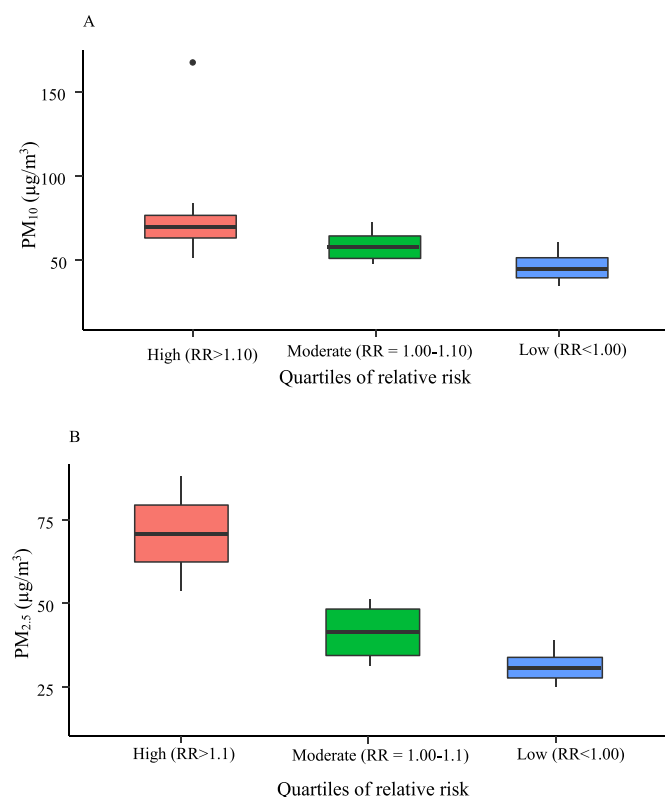


Fig. 2. The relative risks of mortality and associated concentration of PM₁₀ and PM_{2.5}.

studies required to obtain a full symmetry in the Funnel plot ($p < 0.001$). Furthermore, there is no significant change in the results of the sensitivity analysis (Supplementary Figs. S17–S19).

4. Discussion

In this meta-analysis, a significant association between the infant and child under-five mortality and exposure to PM₁₀ and PM_{2.5} were achieved. Briefly, the pooled estimated effect size for PM₁₀ and PM_{2.5} based on 10 µg/m³ increment with infant and child under-five mortality were 1.025 and 1.034, respectively. Several consistent reports have documented the significant association between particulate matter with mortality (Barnett et al., 2005; Bates, 1995; Clemente et al., 2017; Currie and Neidell, 2005; Goldizen et al., 2016; Kim, 2004; Loomis et al., 1999; Nhung et al., 2018; Salvi, 2007; Sierra-Heredia et al., 2018). The estimated risks of mortality and exposure to ambient PM₁₀ from these studies were between 1.01 and 1.04 (Barnett et al., 2005; Bates, 1995; Clemente et al., 2017; Currie and Neidell, 2005; Goldizen et al., 2016; Kim, 2004; Loomis et al., 1999; Nhung et al., 2018; Salvi, 2007; Sierra-Heredia et al., 2018). Generally, exposure to PM_{2.5} reported stronger associations with mortality than PM₁₀. PM_{2.5} can be deposited in the alveoli, and it could pass to the systemic circulation (Zhang et al., 2016). Hence, PM_{2.5} is associated with a wide range of diseases (Alvarado-Cruz et al., 2018; Fu et al., 2019; Li et al., 2017b; Wagner and Zota, 2018; Zhao et al., 2017). The highest causal relations are established between PM_{2.5} and pulmonary and cardiovascular disease (Pun et al., 2017; Sinkemani et al., 2018). Particular causal relations have been recognized between PM_{2.5} and cardiovascular mortality, myocardial infarction, and hypertension (Li et al., 2017a; Lin et al., 2017; Pun et al., 2017; Zhao et al., 2017). Causal associations have also been recognized between PM_{2.5} and lung cancer and chronic obstructive pulmonary disease (COPD) (Ghazipura et al., 2019; Han et al., 2019). The International Agency for Research on Cancer (IARC)

has reported that ambient air pollution and airborne particulate matter are proven group 1 human carcinogens (Loomis et al., 2013). Moreover, oxidative stress and inflammation mechanisms play an important role in respiratory mortality via exposure to airborne particulate (Anderson et al., 2012).

Our findings show the positive association between infant and child under-five mortality with gaseous air pollutants. These findings are consistent with previous observations (Barnett et al., 2005; Bates, 1995; Clemente et al., 2017; Currie and Neidell, 2005; Goldizen et al., 2016; Kim, 2004; Loomis et al., 1999; Nhung et al., 2018; Salvi, 2007; Sierra-Heredia et al., 2018). The percent change of infant and child mortality with exposure to NO₂, CO, and O₃ from these studies were between -8–16%, 0–15%, and -10.8–27%, respectively (Barnett et al., 2005; Bates, 1995; Clemente et al., 2017; Currie and Neidell, 2005; Goldizen et al., 2016; Kim, 2004; Nhung et al., 2018; Salvi, 2007; Sierra-Heredia et al., 2018). SO₂ possibly triggers bronchoconstriction and decrements the lung function. Previous studies have suggested that SO₂ at low concentrations has stronger effects on mortality (Ren et al., 2017; Tanaka, 2015), which is in agreement with our findings. In Wuhan, Ren et al. (2017) found that daily respiratory mortality increased with SO₂ by 2.4% per 10 µg/m³ increment in SO₂ concentration (Ren et al., 2017). Tanaka et al. (2015) in 175 Chinese cities found that reduction of industrial emission, particularly SO₂ was associated with a 20%–63% decrease in neonatal and infant mortality (Tanaka, 2015). SO₂ due to its high-water solubility is affecting the noise and upper respiratory system (Ren et al., 2017). SO₂ mainly comes from the burning of fossil fuels.

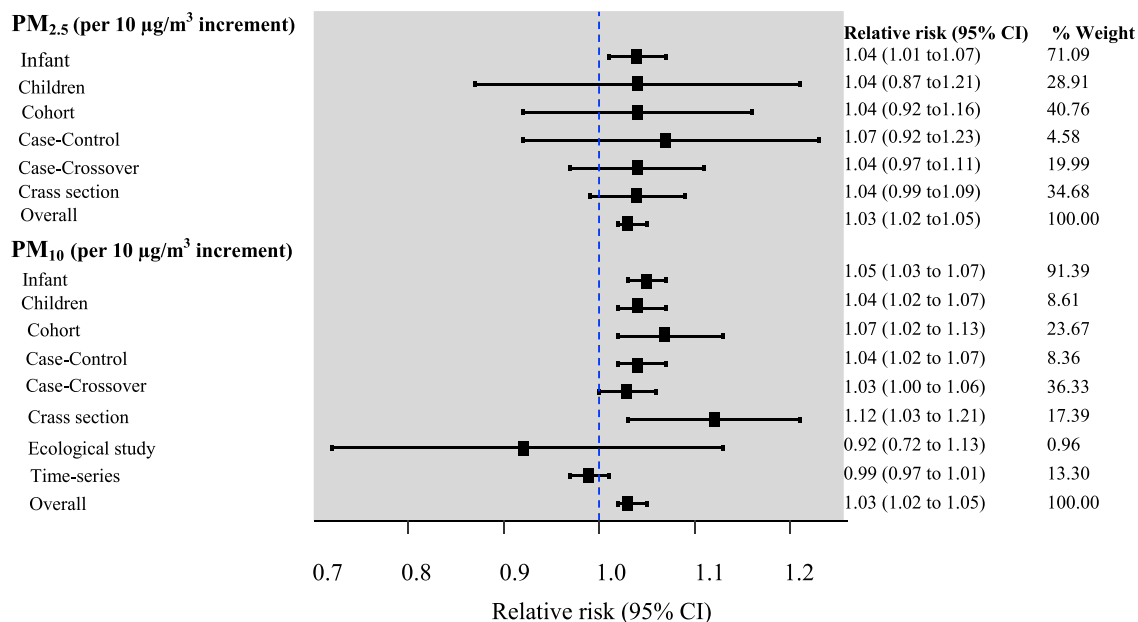
NO₂ as a precursor of O₃, mainly comes from motor vehicle emissions. Exposure to NO₂ has been associated with impairment of lung function, bronchial inflammation (Shah et al., 2015) and increases in respiratory illness as well as increased susceptibility to airway infections (Kelly et al., 1996). Crouse et al. (2015) found a positive association with NO₂ and mortality from respiratory diseases (Crouse et al., 2015). Another study by Vanos et al. (2014) in 10 Canadian cities show the strongest associations between NO₂ (13.3%) and CO (10.7%) exposure with respiratory mortality (Vanos et al., 2014).

We found no evidence of associations between O₃ concentrations and the risk of infant and child deaths. This results consistent with the funding of two meta-analyses conducted by <https://ehp.niehs.nih.gov/doi/full/10.1289/ehp.9014> Bell et al. (2005) and Anderson et al. (2004). The study conducted by Goudarzi et al. (2015) showed that O₃ was significantly associated with respiratory mortality (Goudarzi et al., 2015). Airway inflammation and airway hyper-responsiveness are the two main causes of O₃ mortality. But, in some other studies not observe a significant effect of O₃ on respiratory mortality (Dadbakhsh et al., 2015). Generally, studies that used peak O₃ concentrations measured during the warmer months reported positive associations. The specific causal relations between ambient air pollution and infant and child mortality have reviewed by WHO (Drisse, 2018).

4.1. Mortality mechanisms of air pollution

The exact underlying mechanisms of the air pollution effects on the infant and under-five deaths are not completely understood, but the interplay of environmental influences and epigenetic factors are important (Korten et al., 2017; Miller and Ho, 2008). The immune system and lungs of the child during the first few years of life are extremely immature. Hence, the child and infants compared to adults are more vulnerable to air pollutants (Deng et al., 2016; Gouveia et al., 2018). Moreover, another special vulnerability of the infant and child to air pollutants are associated with high rates of respiratory disease and incomplete metabolic systems, as well as the immature host defenses (Glinianaia et al., 2004; Ha et al., 2003). On the other hand, adverse health effects of air pollution on the cardiovascular and respiratory system, such as oxidative stress, inflammation, impaired coagulation, and endothelial function play an important role in mortality (Kannan et al., 2006). In this regard, numerous mortality risk factors from

A



B

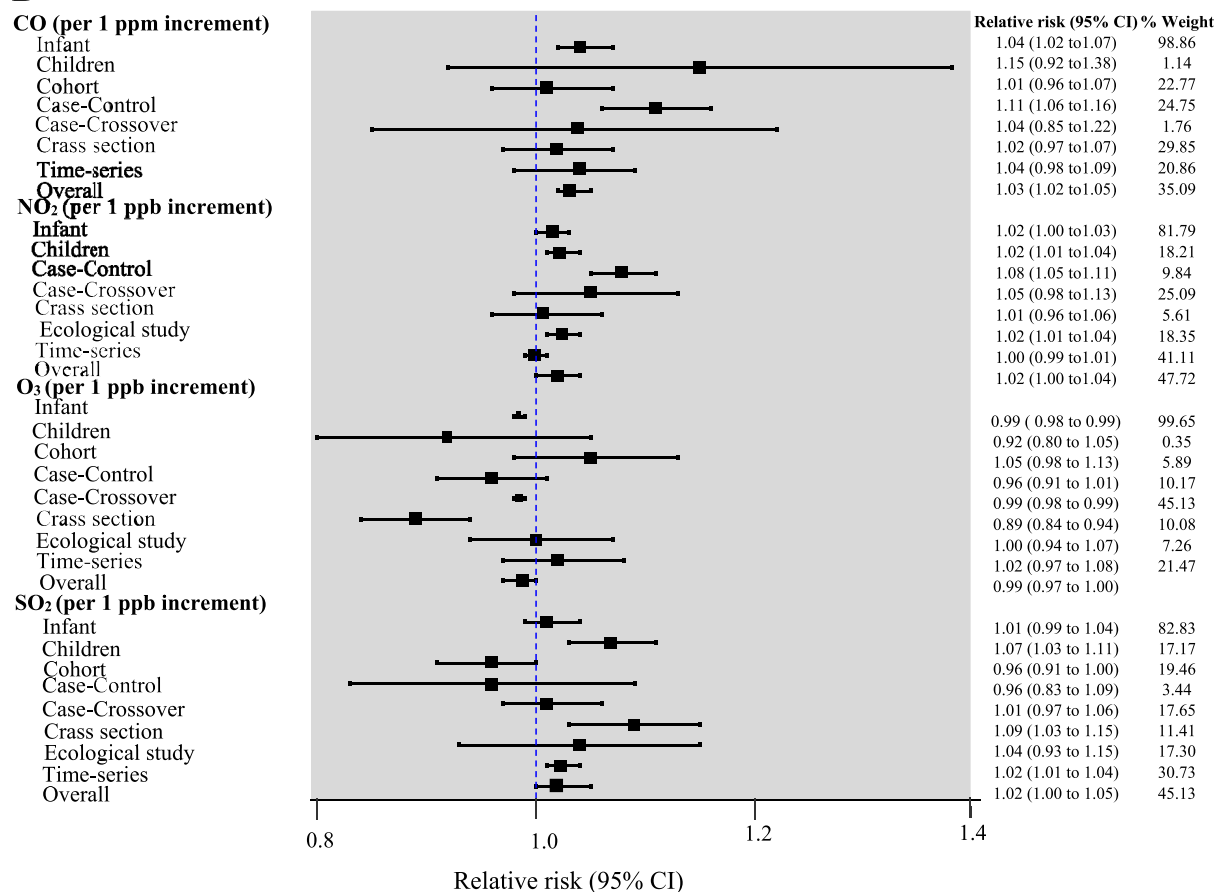


Fig. 3. Relative risk (with 95% CI) of infant and child under 5-years mortality with exposure to the particulate (a) and gaseous (b) stratified by age (infant or child), and design of the study.

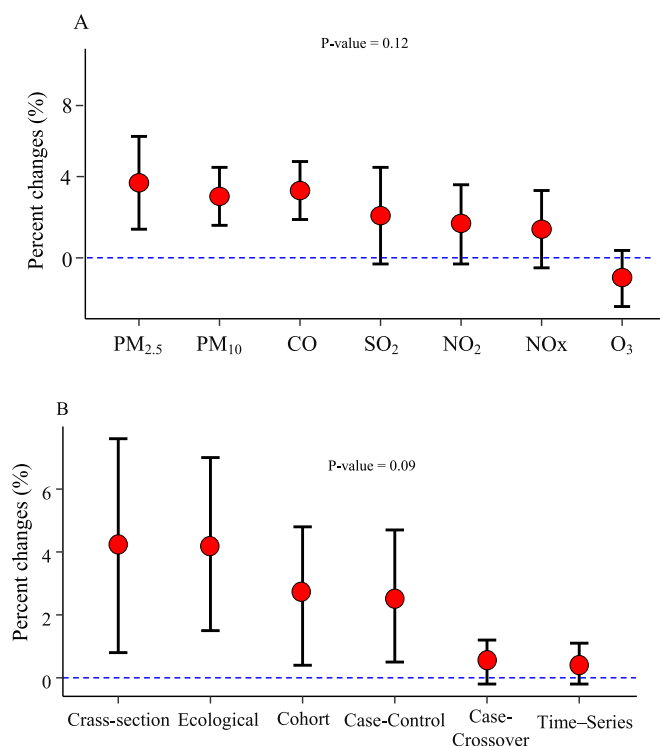


Fig. 4. Pooled percent changes (with 95% CI) of infant and child under 5-years mortality with exposure to air pollutants (a) and type of study (b).

respiratory and cardiovascular disease are associated with air pollution including acute myocardial infarction, cardiac arrhythmias, and progression of atherosclerosis (Landrigan et al., 2018).

The current study had several remarkable strengths. This meta-analysis combined and synthesis all existing data concerning infant and child mortality and air pollution. The results obtained from the current study could be applied to risk assessments of infant and child mortality with exposure to ambient air pollution. We also checked any evidence of study biases by examining funnel plots with Egger and Begg's tests, as well as sensitivity analyse to assess the impact of each study on summary estimates. However, evidence of the infant and child mortality with age group and prenatal exposure effects are still insufficient for meta-analysis and uncertainty can be generated when converting an OR to a RR.

5. Conclusions

The results of the current meta-analysis provide clear evidence that PM_{2.5}, PM₁₀, CO, SO₂ and NO₂ were significantly associated with the infant and child under-five mortality. All investigated air pollutants were associated with increased risk of infant and child under-five mortality, but the effects of PM_{2.5} and PM₁₀ were stronger. So, recognizing the sources and concentrations of air pollutants will help to establish the new regulations for control measures. This subsequently leads to the diminishing of mortality of infant and children under age of five. Moreover, further cohort studies in this field are required to be conducted to better investigate the effects of air pollution on health effects.

Declaration of competing interest

All of the authors declare that they have no actual or potential of conflict of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.apr.2020.02.006>.

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